

Course Code: CS-307	Course Name: Computer Networks
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Student Roll No:	Section No:

Instructions:

- Return the question paper.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- All questions carry 10 marks.
- In case of any ambiguity, you may make an assumption. But your assumption should not contradict any statement in the question paper.

Time: 60 minutes.

Question 1: Consider the scenario shown in figure 1, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 100$ Mbps. The four links from the servers to the shared link have a transmission capacity of $R_S = 20$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 90$ Mbps per second. You might want to review Figure 1.20 in the text before answering the following questions:

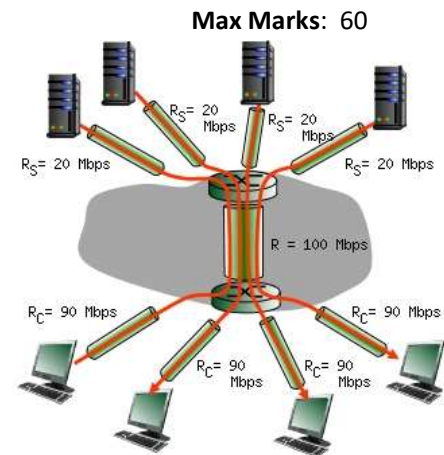


Figure 1

- What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fair-shared (i.e., divides its transmission rate equally among the four pairs)?
- Which link is the bottleneck link for each session?
- Assuming that the senders are sending at the maximum rate possible, what are the link utilizations for the sender links (R_S), client links (R_C), and the middle link (R)?

Question 2: Consider the network setup shown in figure 2 with two end-hosts and three routers R_1 , R_2 , and R_3 (i.e. $N=3$). Now suppose that the web server sends 20 HTTP responses each of size 20K to the client. Assume any missing values but do explicitly mention them as part of your answer.

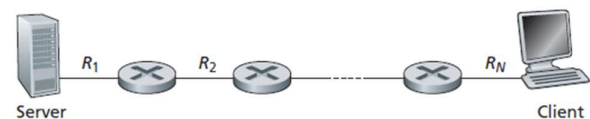


Figure 2

- How many TCP connections will be created for exchanging 20 responses?
- How many times transport layers will be activated? And where?
- How many times network and data link layers will be activated? And where?
- Write the name and type of physical mediums of each link shown.

Question 3: Suppose that the Alice as a client wants to retrieve the www.google.com page but has no information about the www.google.com web server IP address:

- Describe the process of the Alice client obtaining the IP address for the hostname www.google.com under the assumption that it is not cached at the local DNS server and that the local DNS server has not cached an entry for the .com DNS server. (Describe this for the non-recursive case)
- What transport protocol(s) does DNS use: TCP, UDP, or both?
- How many types of Resource Records (RR) are there?
- Is it possible for an organization's web server and mail server to have exactly the same alias for a hostname (for example, foo.com)? What would be the type for the RR that contains the hostname of the mail server?

Question 4: Suppose Alice, with a Web-based e-mail account (such as Hotmail or Gmail), sends a message to Bob, who accesses his mail from his mail server using IMAP. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts.

Question 5: The text below shows the reply sent from the server in response to the HTTP GET message. Answer the following questions, indicating where in the message you find the answer.

```
HTTP/1.1 200 OK<cr><lf>Date: Tue, 07 Mar 2008 12:39:45GMT<cr><lf>Server: Apache/2.0.52 (Fedora)
<cr><lf>Last-Modified: Sat, 10 Dec2005 18:27:46 GMT<cr><lf>ETag: "526c3-f228a4c80"<cr><lf>Accept-
Ranges: bytes<cr><lf>Content-Length: 3874<cr><lf>Keep-Alive: timeout=max=100<cr><lf>Connection: Keep-
Alive<cr><lf>Content-Type: text/html; charset=ISO-8859-1<cr><lf><cr><lf><!doctype html public "-//w3c//dtd html 4.0
transitional//en"><lf><html><lf><head><lf> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-
1"><lf> <metaname="GENERATOR" content="Mozilla/4.79 [en] (Windows NT5.0; U) Netscape]"><lf> <title>CMPSCI
453 / 591 / NTU-ST550A Spring 2005 homepage</title><lf></head><lf><much more document text following here (not
shown)>
```

- Was the server able to successfully find the document or not? What time was the document reply provided?
- When the document was last modified?
- How many bytes are there in the document being returned?
- What are the first 5 bytes of the document being returned? Did the server agree to a persistent connection?

Question 6: Consider the scenario in the figure 3, for which there is an institutional network connected to the Internet. Suppose that the average object size is 850,000 bits and that the average request rate from the institution's browsers to the origin servers is 16 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average. Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to institution router) and the average Internet delay. For the average access delay, use $\Delta/(1 - \Delta\beta)$, where Δ is the average time required to send an object over the access link and β is the arrival rate of objects to the access link.

- Find the total average response time.
- Now suppose a cache is installed in the institutional LAN with a miss rate of 0.4. Find the total response time.

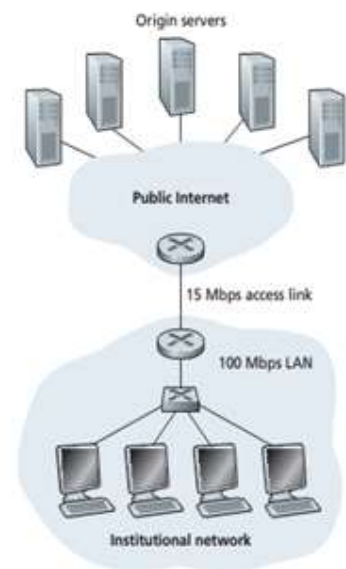


Figure 3

BEST OF LUCK!